

## ***Uricosuric effect of Roselle (*Hibiscus sabdariffa*) in normal and renal-stone former subjects.***



AIM OF THE STUDY: The Roselle (*Hibiscus sabdariffa*) was investigated for its uricosuric effect. MATERIALS AND METHODS: A human model with nine subjects with no history of renal stones (non-renal stone, NS) and nine with a history of renal stones (RS) was used in this study. A cup of tea made from 1.5 g of dry Roselle calyces was provided to subjects twice daily (morning and evening) for 15 days. A clotted blood and two consecutive 24-h urine samples were collected from each subject three times: (1) at baseline (control); (2) on days 14 and 15 during the tea drinking period; and (3) 15 days after the tea drinking was stopped (washout). Serum and 24-h urinary samples were analyzed for uric acid and other chemical compositions related to urinary stone risk factors. RESULTS: All analyzed serum parameters were within normal ranges and similar; between the two groups of subjects and among the three periods. Vis-à-vis the urinary parameters, most of the baseline values for both groups were similar. After taking the tea, the trend was an increase in oxalate and citrate in both groups and uric acid excretion and clearance in the NS group. In the RS group, both uric acid excretion and clearance were significantly increased ( $p < 0.01$ ). When the fractional excretion of uric acid (FEUa) was calculated, the values were clearly increased in both the NS and SF groups after the intake of tea and returned to baseline values in the washout period. These changes were more clearly observed when the data for each subject was presented individually. CONCLUSIONS: Our data demonstrate a uricosuric effect of Roselle calyces. Since the various chemical constituents in Roselle calyces have been identified, the one(s) exerting this uricosuric effect need to be identified.